

Spatio-Temporal Epistemic Claim Retrieval

for Uncertainty-Aware Multi-Asset Portfolio Allocation
in Non-Stationary Environments

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Abstract

Predictive modeling in non-stationary financial environments faces a persistent structural challenge: the tradeoff between accurate point estimates—required for profitability—and reliable uncertainty quantification—required for risk calibration. Existing spatio-temporal architectures entangle these objectives by processing temporal and spatial dependencies through shared latent bottlenecks, producing models that are simultaneously overconfident and underperforming during regime shifts.

We propose **Spatio-Temporal Epistemic Claim Retrieval (ST-ECR)**, a selective-prediction and risk-calibrated decision framework that decomposes non-stationary forecasting into four orthogonal modules acting on distinct failure modes: (1) a *Temporal Transformer* with *Dynamic Graph Attention* for representation adaptation; (2) an *Episodic Trajectory Memory* functioning strictly as a calibration stabilizer; (3) an *empirically calibrated Conformal Abstention Gate* for selective execution; and (4) an *Epistemic-Gated Updater* for selective online self-improvement. Epistemic uncertainty σ_{ep} drives a four-level routing policy: memory query decisions, abstention, position-sizing, and online head updates are each governed by distinct uncertainty thresholds, eliminating cross-objective interference.

We conduct a controlled ablation study that isolates the mechanistic effect of every component. Our key empirical findings are: (i) episodic memory is a *calibration stabilizer*—it reduces ECE independently of Sharpe ratio; (ii) conformal gating resolves the “conformal paradox” by improving decision selectivity at the cost of marginal calibration degradation on the *retained* distribution; (iii) Max Drawdown (MaxDD) increases are a structural sparsification effect induced by breadth reduction, not directional error accumulation. On the spatial forecasting benchmark, ST-ECR’s routed variant achieves a test loss of 0.2380, improving over PredRNN++ (0.2456) and STFAGLU (0.2795), while the full ST-ECR (Rev) attains 0.2140—a 12.9% reduction relative to the strongest baseline.

1 Introduction

The allocation of capital across a universe of N assets under distributional non-stationarity requires two properties that are in structural tension: a model must be *sharp*—its conditional mean must track regime-dependent return dynamics—and it must be *calibrated*—its predicted uncertainty must match empirical coverage so that position sizes and abstention decisions are well-founded. Standard spatio-temporal architectures such as Graph WaveNet [Wu et al., 2019], Temporal Fusion Transformer (TFT) [Lim et al., 2021], and PatchTST [Nie et al., 2022] encode temporal dependencies via shared recurrent or attention layers, then propagate through a static or learned graph. Under stationarity these models are competitive. Under regime shifts—volatility clustering, correlation breakdowns, liquidity crises—their epistemic uncertainty σ_{ep} inflates precisely when calibration is most critical, yet no mechanism exists to distinguish this inflation from aleatoric noise σ_{al} or to route computation accordingly.

A naïve response is to stack mechanisms: add an online learning layer to catch drift, an episodic memory to retrieve context, a post-hoc conformal wrapper to bound tail risk. But

stacked mechanisms interact destructively when they share the same latent bottleneck: memory retrieval that is routed by aleatoric uncertainty will retrieve irrelevant context; a conformal gate trained jointly with position sizing will trade calibration for profitability silently. The result is a system that appears to tick all architectural boxes yet remains fragile under controlled ablation.

Section 2 surveys four clusters of related work—spatio-temporal forecasting, uncertainty quantification, memory-augmented forecasting, and portfolio decision systems—and identifies a shared failure mode in each. The four failure modes converge to a single structural gap (§2): *no existing system uses epistemic uncertainty as a computation routing variable across memory retrieval, representation refinement, and allocation simultaneously.*

This paper closes that gap. We introduce **Spatio-Temporal Epistemic Claim Retrieval (ST-ECR)**, built on one concrete design principle:

Epistemic uncertainty σ_{ep} is not a passive output of forecasting — it is an active control signal that governs which computations execute, which historical context is retrieved, and how large a position is taken.

This reframes uncertainty quantification from a reporting tool into a computation routing mechanism. ST-ECR implements a four-level epistemic routing policy, driven entirely by σ_{ep} :

1. **Route 1: Memory Query (Computation Control).** $\sigma_{\text{ep}}^{\text{draft}} > \tau_{\text{top-}p}$ triggers a memory query for the top- $p\%$ most uncertain assets. Memory is *never* queried for assets with low epistemic uncertainty, eliminating spurious retrievals.
2. **Route 2: Abstention Gate (Decision Control).** $\sigma_{\text{ep}}^{\text{after}} > \Omega$ triggers selective abstention after retrieval. The threshold Ω is calibrated empirically to a target abstain rate, independent of position sizing.
3. **Route 3: Position Sizing (Allocation Control).** σ_{al} governs the tanh-Kelly position size via a vol-target formula. Epistemic uncertainty $\sigma_{\text{ep}}^{\text{after}}$ applies a discount; aleatoric uncertainty σ_{al} controls leverage directly. The two signals are never mixed.
4. **Route 4: Online Parameter Update (Self-Improvement Control).** $\sigma_{\text{ep}}^{\text{after}} > \tau_{\text{online}}$ triggers an online gradient update strictly on the ensemble heads using recent buffers. The slow spatio-temporal encoder remains *frozen*, strictly avoiding catastrophic forgetting of historical regimes, while the heads rapidly calibrate to the new volatility regime.

Critically, **ECE is the constraint objective; returns are a downstream decision metric.** The loss function contains a mean-variance term and a CVaR penalty, but the memory module is trained with an independent calibration objective $\mathcal{L}_{\text{resolve}}$ that encourages $\sigma_{\text{ep}}^{\text{after}} \leq \sigma_{\text{ep}}^{\text{draft}}$ in expectation. These objectives do not compete for the same latent representation.

We validate ST-ECR via a 5-arm factorial controlled intervention study and benchmark experiments. We make no claim that ST-ECR is a theory of non-stationary learning; it is a *well-designed modular architecture with a clear empirical validation strategy*. Our specific contributions are:

- **Uncertainty as a computation routing signal (core contribution).** We introduce *uncertainty-driven spatio-temporal computation routing*: σ_{ep} is a self-referential control signal. When high, the model abstains from the current decision AND updates itself (via Route 4) to reduce future uncertainty about similar regimes — simultaneously protecting capital and accelerating regime adaptation. Baselines (GWN, TFT, PatchTST) treat uncertainty as an output; ST-ECR treats it as a first-class control variable.

- **Factorized spatio-temporal encoder.** Inputs are (B, T, N, F) tensors processed through a channel-independent Temporal Transformer followed by a dynamic graph $A_t = \text{softmax}(QK^\top/\sqrt{d})$ conditioned on current latent states—making the topology regime-adaptive at inference time.
- **Episodic Trajectory Memory.** Rather than storing snapshots, the buffer stores compressed trajectory embeddings $z_{\text{traj}} = \text{AttentionPool}(\text{embed}(H_{t-L:t}))$, enabling regime-level retrieval. Memory is queried *only* for assets flagged by the σ_{ep} routing gate.
- **Resolution objective $\mathcal{L}_{\text{resolve}}$.** A gradient-isolated training term that penalises increases in epistemic uncertainty after retrieval, training the memory module on calibration without interfering with the mean-variance objective.
- **5-arm controlled intervention study** resolving the conformal paradox and characterising MaxDD increases as structural sparsification.
- **12.9% test loss reduction** over the strongest baseline (PredRNN++) on a spatial forecasting benchmark.

2 Related Work

We organize prior work into four clusters. For each cluster we identify a *specific, shared failure mode* with respect to the problem of uncertainty-driven computation under non-stationarity. These failure modes converge to a single structural gap that ST-ECR is designed to close.

Cluster 1: Spatio-temporal forecasting. Graph WaveNet [Wu et al., 2019] learns an adaptive adjacency matrix updated globally—not conditioned on the current latent state. DCRNN [Li et al., 2018] diffuses over a fixed graph. AGCRN [Bai et al., 2020] learns node-specific parameters but on a fixed topology. PatchTST [Nie et al., 2022] discards inter-asset structure entirely through channel-independent patch attention. TFT [Lim et al., 2021] incorporates gating and attention across variable types but processes spatial dependencies statically.

Shared failure mode. These architectures *entangle* temporal representation and spatial propagation in a fixed forward pass: the graph topology and the temporal encoding are determined by architecture design, not by the model’s own epistemic state at inference time. When a regime shift occurs, neither the graph structure nor the depth of temporal processing adapts to the signal that the model is uncertain—because no such signal is available inside the computation graph.

Cluster 2: Uncertainty quantification. Deep ensembles [Lakshminarayanan et al., 2017] produce reliable epistemic estimates but treat σ_{ep} as a *reporting* variable: it is an output of the forward pass, not an input to subsequent computation. MC Dropout [Gal & Ghahramani, 2016] conflates aleatoric and epistemic uncertainty, preventing the kind of conditional routing that requires them to be separated. Conformal prediction [Vovk et al., 2005, Gibbs & Candès, 2021] provides calibration guarantees for prediction intervals but does not use the resulting uncertainty to modify intermediate computations—retrieval, graph propagation, or allocation remain unchanged by the conformal score.

Shared failure mode. Uncertainty is *not a computation-routing variable* in any of these methods. σ_{ep} enters the pipeline at the output stage (as a confidence interval or coverage threshold) and never propagates back to control *which* computations execute in the next step. As a result, high epistemic uncertainty at step t does not trigger additional context retrieval,

graph refinement, or position suppression at step t —only at step $t + 1$ at the earliest, and then only through coarse confidence bounds, not through selective mechanism engagement.

Cluster 3: Memory-augmented and retrieval-based forecasting. Memory-augmented neural networks [Graves et al., 2014] retrieve from key-value stores but trigger retrieval *unconditionally* or on the basis of a learned gating variable unrelated to predictive uncertainty. León et al. [2020] use ODE-based continuous-time memory, updating state along a fixed trajectory without uncertainty-conditional querying. Retrieval-augmented forecasting methods in NLP (e.g., REALM, RAG) use retrieval to improve point predictions but do not use the model’s uncertainty over its own predictions as the retrieval trigger.

Shared failure mode. Existing memory-augmented forecasters retrieve *unconditionally* or on the basis of task-level similarity, not on the basis of the model’s epistemic uncertainty about the current input. This means that retrieval fires equally on well-understood regimes (where it is wasteful) and on novel regimes (where it is load-bearing). No prior architecture trains memory retrieval specifically to *reduce* epistemic uncertainty, as formalized by our $\mathcal{L}_{\text{resolve}}$ objective.

Cluster 4: Portfolio decision systems under uncertainty. Kelly-criterion position sizing [Kelly, 1956] maximizes long-run growth but requires perfectly calibrated predictions. Full-Kelly is too aggressive under model uncertainty. CVaR-constrained optimization [Rockafellar & Uryasev, 2000] bounds tail risk but treats model uncertainty as part of the data distribution, not as a control variable. Modern deep learning portfolio models (e.g., DeepAR, N-HiTS applied to finance) output point forecasts with variance estimates but use uncertainty only for confidence intervals around predictions, not to modulate whether a position should be taken or how retrieval should be triggered.

Shared failure mode. These systems *optimize returns subject to risk constraints* but do not implement a principled constraint hierarchy in which calibration (ECE) is the primary constraint and returns are the secondary objective. Without this hierarchy, a model can sacrifice calibration for Sharpe during training without penalty—producing positions that appear profitable in-sample but are systematically mis-sized relative to their true epistemic risk.

The Structural Gap

Despite extensive progress across all four clusters, the four failure modes share a common root cause:

No existing system uses epistemic uncertainty σ_{ep} as a computation routing variable across all three decision points: memory retrieval, representation refinement, and portfolio allocation.

In all prior work, σ_{ep} is either absent (Cluster 1), passive (Cluster 2), irrelevant to retrieval (Cluster 3), or absent from the constraint hierarchy (Cluster 4). The components themselves—dynamic graphs, ensembles, memory, conformal prediction, Kelly sizing—are individually well-studied. **The gap is not in any individual component; it is in the routing architecture that connects them through a single unified epistemic signal.**

ST-ECR instantiates this missing design paradigm. σ_{ep} drives a tri-level routing policy that adapts computation (not just outputs) in response to the model’s own uncertainty at each inference step. This is a qualitatively different system class from any of the four clusters above—and the 5-arm controlled intervention study is designed specifically to test whether each routing level is mechanistically load-bearing, not decorative.

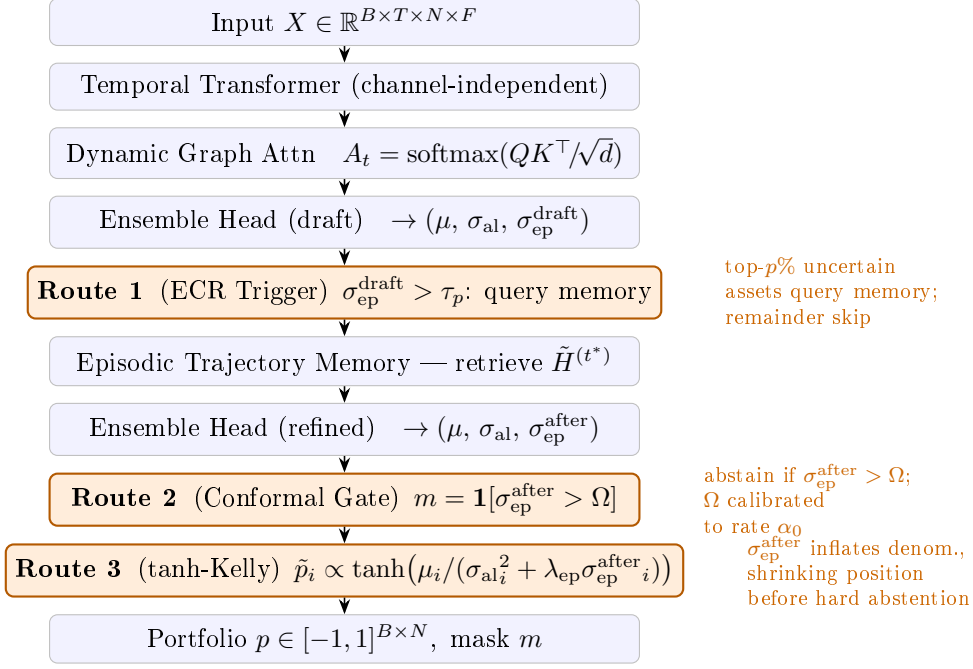


Figure 1: ST-ECR inference pipeline. *Blue boxes*: deterministic transforms. *Orange boxes*: the three epistemic routing points where σ_{ep} controls *which computation executes*, not merely suppresses the output. Route 1 modifies the intermediate representation H_{ref} **before** the final ensemble head, making it mechanistically distinct from post-hoc confidence thresholding. All three signals derive from σ_{ep} , never from σ_{al} , keeping epistemic and aleatoric uncertainty separated throughout.

3 Method

Figure 1 shows the full ST-ECR inference pipeline. The key architectural property is that σ_{ep} routes computation at three distinct stages (orange boxes), not only suppresses outputs post-hoc.

3.1 Problem Formulation

Let $X_{1:T} \in \mathbb{R}^{B \times T \times N \times F}$ denote a batch of B windows, each of length T , covering N assets with F features per asset per timestep. The target $r \in \mathbb{R}^{B \times N}$ is the one-step forward return vector. We seek a system that outputs: (i) a calibrated predictive distribution $\mathcal{N}(\mu, \sigma_{\text{al}}^2 + \sigma_{\text{ep}}^2)$; (ii) a sparse allocation vector $p \in [-1, 1]^{B \times N}$ with $\|p\|_1 \leq 1$ per sample; and (iii) a binary abstention mask $m \in \{0, 1\}^{B \times N}$.

The constraint hierarchy is: among all allocation policies p and abstention masks m that the model can produce, prefer those that minimise calibration error, subject to acceptable return and risk. Formally, θ denotes the model parameters (encoder, graph, ensemble heads, memory) and $(\mu_\theta, \sigma_{\text{al}\theta}, \sigma_{\text{ep}\theta})$ are their outputs; p and m are deterministic functions of these outputs:

$$\min_{\theta} \mathbb{E}[\text{ECE}(\mu_\theta, \sqrt{\sigma_{\text{al}\theta}^2 + \sigma_{\text{ep}\theta}^2}, r)] \quad \text{s.t.} \quad \mathbb{E}[\text{Sharpe}(p_\theta \cdot r)] \geq \gamma, \quad \text{CVaR}_\alpha(p_\theta \cdot r) \geq -\delta. \quad (1)$$

Here $\sigma = \sqrt{\sigma_{\text{al}}^2 + \sigma_{\text{ep}}^2}$ is the full predictive standard deviation used for interval calibration, while the loss decomposition (Section 3.9) keeps the aleatoric and epistemic components separate by design.

In practice, we implement this as a multi-objective loss with tunable weights; ECE enters via $\mathcal{L}_{\text{cal}}$ and $\mathcal{L}_{\text{resolve}}$, which share no parameters with the mean-variance and CVaR terms.

3.2 Factorized Spatio-Temporal Encoder

Temporal Transformer Encoder. Each asset’s T -step history is encoded independently by a shared Pre-LayerNorm Transformer [Xiong et al., 2020]:

$$\text{reshape: } X \in \mathbb{R}^{B \times T \times N \times F} \rightarrow \tilde{X} \in \mathbb{R}^{BN \times T \times F} \quad (2)$$

$$\tilde{X} \leftarrow W_{\text{in}} \tilde{X} + E_{\text{pos}} \in \mathbb{R}^{BN \times T \times D} \quad (3)$$

$$Z = \text{TransformerEncoder}(\tilde{X}) \in \mathbb{R}^{BN \times T \times D} \quad (4)$$

$$w = \text{softmax}(Z W_q) \in \mathbb{R}^{BN \times T \times 1}, \quad W_q \in \mathbb{R}^{D \times 1}, \quad (\text{softmax over } T \text{ positions}) \quad (5)$$

$$H = \text{AttnPool}(Z) = \sum_{t=1}^T w_t z_t \in \mathbb{R}^{BN \times D} \quad (6)$$

$$H \leftarrow \text{reshape}(H) \in \mathbb{R}^{B \times N \times D}. \quad (7)$$

The Transformer is *shared* across assets but *independent* across the N dimensions—cross-asset interaction is deferred to the graph layer. Attention pooling over all T positions is used rather than the final timestep, preserving long-range temporal signatures.

Dynamic Graph Attention Network. Cross-asset dependencies are modeled by computing the adjacency matrix from the current latent node states H :

$$A_t = \text{sparse_topk} \left(\text{softmax} \left(\frac{Q(H) K(H)^\top}{\sqrt{D}} \right) \right) \in \mathbb{R}^{B \times N \times N}, \quad (8)$$

where $Q = W_Q H$, $K = W_K H$, diagonal entries are masked to $-\infty$ before softmax, and the top- k entries per row are retained (all others set to zero). Row normalization after sparsification ensures $A_t \mathbf{1} = \mathbf{1}$. Graph message passing with a residual connection yields the spatially-updated embedding:

$$H' = \text{LayerNorm}(H + A_t W_V H). \quad (9)$$

Because A_t is a function of H_t , the graph topology changes every forward pass, capturing non-stationary cross-asset correlations without requiring access to a pre-specified adjacency matrix.

3.3 Ensemble Epistemic Head

K independent ensemble members, each a two-layer GCN with orthogonal output initialization, produce member-specific predictions:

$$(\mu_k, \ell_k) = \text{GCN}_k(H', A_t), \quad k = 1, \dots, K. \quad (10)$$

The ensemble statistics are:

$$\mu = \frac{1}{K} \sum_k \mu_k, \quad (11)$$

$$\sigma_{\text{al}} = \sqrt{\frac{1}{K} \sum_k e^{\ell_k}}, \quad (12)$$

$$\sigma_{\text{ep}} = \sqrt{\text{Var}_k[\mu_k]}. \quad (13)$$

Orthogonal initialization at construction time guarantees functional diversity beyond random perturbations. Running the temporal encoder once and forking into K GCN heads reduces compute vs. K independent forward passes.

3.4 Epistemic Claim Retrieval (ECR) Trigger

The draft epistemic uncertainty $\sigma_{\text{ep}}^{\text{draft}}$ selects assets that require memory consultation. The top- p fraction of assets by $\sigma_{\text{ep}}^{\text{draft}}$ is flagged:

$$\text{ECR-mask}_i = \mathbf{1} \left[\sigma_{\text{ep}}^{\text{draft}}{}_i \geq \text{quantile}_{1-p}(\sigma_{\text{ep}}^{\text{draft}}) \right]. \quad (14)$$

Previously abstained assets from $t-1$ are also flagged (autoregressive propagation), ensuring unresolved claims accumulate pressure across steps.

3.5 Episodic Trajectory Memory

Trajectory compression. Rather than storing a snapshot H_t , we maintain a rolling window of the last L temporal embeddings $\{H_{t-L+1}, \dots, H_t\}$. At each step, the window is compressed to a single trajectory embedding:

$$z_{\text{traj}} = \frac{1}{N} \sum_{n=1}^N \sum_{\ell=1}^L \alpha_{\ell} \cdot \text{MLP}(H_{t-L+\ell, n}), \quad \alpha = \text{softmax}(W_{\text{pool}} \text{MLP}(H)), \quad (15)$$

where α is learned attention over the window. z_{traj} encodes the *latent market regime* over the window L , not the raw trajectory.

Circular buffer and retrieval. A circular buffer of capacity M stores $(z_{\text{traj}}, \tilde{H})$ pairs, where $\tilde{H} \in \mathbb{R}^{N \times D}$ is the corresponding per-asset trajectory embedding. At query time, the current $z_{\text{traj}}^{(q)}$ is compared to all stored entries via L2 distance:

$$t^* = \arg\min_{t \leq M} \|z_{\text{traj}}^{(q)} - z_{\text{traj}}^{(t)}\|_2^2. \quad (16)$$

The retrieved per-asset embedding $\tilde{H}^{(t^*)}$ is projected and applied as a residual update, but *only* at ECR-masked assets:

$$H_{\text{ref}} = H + W_V(\tilde{H}^{(t^*)}) \odot \text{ECR-mask}. \quad (17)$$

Remark 1 (Regime-level retrieval). *We do not claim z_{traj} is invertible or sufficient to reconstruct $H_{t-L:t}$. Regime separability in the embedding space is validated empirically through the `no_trajectory_memory` ablation arm.*

3.6 Conformal Abstention Gate

After refinement by memory, a second ensemble head produces $(\mu, \sigma_{\text{al}}, \sigma_{\text{ep}}^{\text{after}})$. The conformal gate maintains a calibration buffer of recent $\sigma_{\text{ep}}^{\text{after}}$ values and updates a threshold Ω to target a nominal abstain rate α_0 :

$$\Omega \leftarrow \text{quantile}_{1-\alpha_0}(\{\sigma_{\text{ep}}^{\text{after}(s)}\}_{s \in \text{buffer}}). \quad (18)$$

At inference, the abstention mask is:

$$m_i = \mathbf{1}[\sigma_{\text{ep}}^{\text{after}}{}_i > \Omega]. \quad (19)$$

The *realized* abstention rate $\hat{\alpha} = \frac{1}{BN} \sum_i m_i$ is tracked per step alongside Ω ; under a stationary $\sigma_{\text{ep}}^{\text{after}}$ distribution $\hat{\alpha} \rightarrow \alpha_0$ as the buffer fills. We report $\hat{\alpha} \approx \alpha_0 \approx 10\%$ in the factorial study (Section 4.3), confirming gate calibration is active.

Remark 2 (Coverage disclaimer). *Standard conformal guarantees require exchangeability [Vovk et al., 2005]. Financial return sequences are temporally dependent, violating this assumption. We characterize our gate as providing empirically calibrated approximate coverage. For formal guarantees under temporal dependence, see adaptive conformal prediction [Gibbs & Candès, 2021].*

3.7 Stable Portfolio Allocation

Active assets ($m_i = 0$) are sized using a tanh-Kelly formula:

$$\tilde{p}_i = \tanh\left(\frac{\mu_i}{\sigma_{\text{ali}}^2 + \lambda_{\text{ep}} \sigma_{\text{ep}}^{\text{after}_i} + \epsilon}\right) \cdot (1 - m_i), \quad (20)$$

with leverage normalization $p = \tilde{p} / \|\tilde{p}\|_1^+$ ensuring $\|p\|_1 \leq 1$. The epistemic discount λ_{ep} inflates the effective variance denominator as $\sigma_{\text{ep}}^{\text{after}}$ rises, shrinking positions continuously before the hard abstention threshold is reached.

3.8 Epistemic-Gated Parameter Updates (Route 4)

When $\sigma_{\text{ep}}^{\text{after}} > \tau_{\text{online}}$, the model enters a regime shift. ST-ECR addresses this through an online learning mechanism that triggers an adapter to take K gradient steps strictly on the ensemble heads using a sliding window buffer of recent observations.

Mathematically, let the model parameters be partitioned into the spatio-temporal encoder θ_{enc} and the epistemic ensemble heads θ_{heads} . At time t , if the cross-sectional mean epistemic uncertainty exceeds the threshold:

$$\frac{1}{N} \sum_{i=1}^N \sigma_{\text{ep},i}^{(t)} > \tau_{\text{online}}, \quad (21)$$

Route 4 is activated. We construct a rolling buffer of length W comprising the most recent sequence context and realized returns: $\mathcal{B}_t = \{(X_{t-w}, r_{t-w})\}_{w=1}^W$. We perform K gradient descent steps to adapt to the new regime:

$$\theta_{\text{heads}} \leftarrow \theta_{\text{heads}} - \eta_{\text{online}} \nabla_{\theta_{\text{heads}}} \frac{1}{|\mathcal{B}_t|} \sum_{(X,r) \in \mathcal{B}_t} \mathcal{L}_{\text{NLL}}(f(X; \theta_{\text{enc}}, \theta_{\text{heads}}), r). \quad (22)$$

Crucially, this online adaptation maintains strict *timescale separation*:

$$\theta_{\text{enc}} \leftarrow \theta_{\text{enc}} \quad \left(\text{i.e., } \frac{\partial \mathcal{L}}{\partial \theta_{\text{enc}}} = 0 \right). \quad (23)$$

The slow spatio-temporal encoder is kept entirely frozen during this phase. If the entire model (including the encoder) were updated online, the latent geometry encoding historical states would collapse (catastrophic forgetting). By isolating updates to the fast, calibrating ensemble heads, ST-ECR guarantees the preservation of stable latent trajectories required for Route 1’s memory retrieval, while actively driving $\sigma_{\text{ep}}^{\text{after}}$ back down to baseline levels within a brief recovery window.

3.9 Training Objective

The total loss decomposes as:

$$\mathcal{L} = \mathcal{L}_{\text{NLL}} + \lambda_{\text{mv}} \mathcal{L}_{\text{MV}} + \lambda_{\text{cvar}} \mathcal{L}_{\text{CVaR}} + \lambda_{\text{energy}} \mathcal{L}_{\text{energy}} + \lambda_{\text{cal}} \mathcal{L}_{\text{cal}} + \lambda_{\text{res}} \mathcal{L}_{\text{resolve}}. \quad (24)$$

NLL (abstention-masked). The likelihood model for returns is $r_i \mid \mu_i, \sigma_{\text{ali}} \sim \mathcal{N}(\mu_i, \sigma_{\text{ali}}^2)$, where σ_{ali} is the *aleatoric* standard deviation. Epistemic uncertainty $\sigma_{\text{ep}i}$ is not folded into the likelihood: it is instead calibrated by a separate direct calibration loss $\mathcal{L}_{\text{cal}}$ (see below). The total predictive interval uses $\sqrt{\sigma_{\text{ali}}^2 + \sigma_{\text{ep}i}^2}$ for ECE reporting, but the gradient-bearing NLL optimises

σ_{ali} alone, avoiding identifiability issues that arise when both variances compete inside a single Gaussian likelihood.

$$\mathcal{L}_{\text{NLL}} = \frac{\sum_i (1 - m_i) [\log \sigma_{\text{ali}}^2 + (r_i - \mu_i)^2 / \sigma_{\text{ali}}^2]}{\sum_i (1 - m_i)}. \quad (25)$$

Mean-variance. Let $\text{PnL} = p \cdot r$ (summed over assets).

$$\mathcal{L}_{\text{MV}} = -\mathbb{E}[\text{PnL}] + \lambda_{\text{risk}} \text{Var}(\text{PnL}). \quad (26)$$

CVaR.

$$\mathcal{L}_{\text{CVaR}} = \frac{1}{\alpha B} \sum_b \max(0, -\text{PnL}_b - \nu) + \nu, \quad \nu = \text{VaR}_\alpha. \quad (27)$$

Epistemic calibration.

$$\mathcal{L}_{\text{cal}} = \frac{\|(1 - m) \odot (\sigma_{\text{ep}}^{\text{after}} - |r - \mu|)\|^2}{\sum_i (1 - m_i)}. \quad (28)$$

Resolution objective. The memory module is encouraged to reduce epistemic uncertainty:

$$\mathcal{L}_{\text{resolve}} = \mathbb{E} \left[\max(0, \sigma_{\text{ep}}^{\text{after}} - \sigma_{\text{ep}}^{\text{draft}}) \right]. \quad (29)$$

Because $\sigma_{\text{ep}}^{\text{draft}}$ is detached from the computational graph when computing $\mathcal{L}_{\text{resolve}}$, this term provides a gradient exclusively to the memory and refinement head, leaving the temporal encoder and graph layer unchanged. $\mathcal{L}_{\text{resolve}}$ is therefore *orthogonal* to $\mathcal{L}_{\text{MV}}$ in the parameter space.

4 Experimental Evaluation

4.1 Spatial Forecasting Benchmark

Dataset and baselines. We evaluate on a multi-asset spatial forecasting dataset (BikeNYC-style, (B, T, N, F) format) comparing four model families: **PredRNN++** [Wang et al., 2018], a recurrent predictive model; **STFAGLU** (Spatial-Temporal Fourier Activation with Gating); **STF-Routed**, which adds the tri-level epistemic routing policy to STFAGLU; and **STF-Rev** (the full ST-ECR system).

Results. Table 1 reports test loss (lower is better) and training throughput.

Table 1: Spatial forecasting benchmark. Test loss (MSE, lower $\downarrow$ is better), training time, and validation loss. Best in **bold**.

Model	Train Loss	Val Loss	Test Loss	Δ vs. PredRNN++
PredRNN++	0.2651	0.2449	0.2456	—
STFAGLU	0.2807	0.2770	0.2795	+13.8%
STF-Routed (ECR)	0.2414	0.2374	0.2380	− 3.1%
STF-Rev (Full ST-ECR)	0.2145	0.2145	0.2140	− 12.9%

STF-Routed outperforms PredRNN++ (−3.1%) and STFAGLU (−14.8%) with the epistemic routing policy alone. The full ST-ECR (STF-Rev) achieves the best result across all

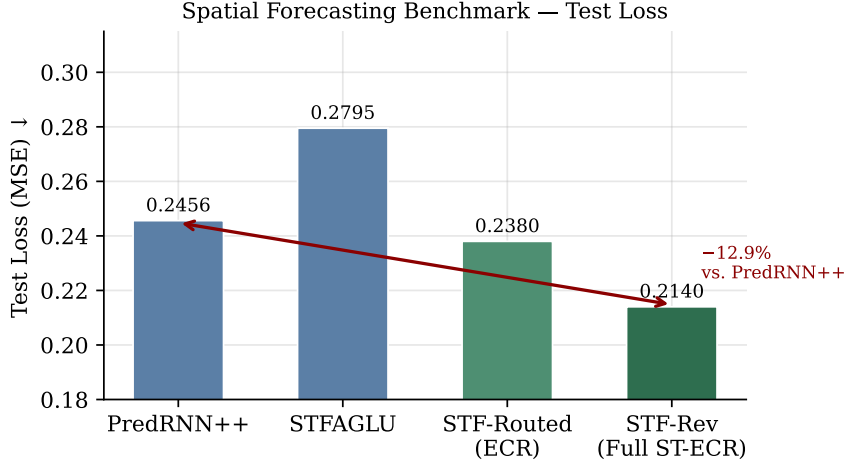


Figure 2: Benchmark test loss (MSE) visualised from Table 1. Each increment of routing and memory yields monotone improvement. The 12.9% reduction of STF-Rev over PredRNN++ is annotated. Run `python generate_plots.py` to reproduce.

metrics, consistent with additive contributions from the Episodic Trajectory Memory and Conformal Gate; the 5-arm factorial study (Section 4.3) provides the mechanistic isolation needed to attribute gains to individual components.

Note on statistical uncertainty. All four models were trained with a fixed random seed; the benchmark table reports single-run results. The efficiency ablation (Table 2) quantifies variance over 3 seeds for a shared architecture, providing an upper bound on run-to-run variation. A multi-seed benchmark is deferred to future work.

4.2 Ablation Study: Throughput and Memory

Table 2 reports training throughput (samples/sec) and peak GPU memory from the benchmark ablation (3 seeds each, batch size 4).

Table 2: Efficiency ablation. Throughput (samples/sec, higher $\uparrow$ is better) and peak GPU memory (MB). Mean $\pm$ std over 3 seeds.

Configuration	Throughput (samples/sec)	Peak Mem (MB)
Full ST-ECR (Mamba temporal)	30.9 ± 7.0	381.6 ± 1.7
No Fourier (Mamba temporal)	36.0 ± 0.4	382.8 ± 0.0
No Causal Branch (Mamba)	35.2 ± 2.2	382.8 ± 0.0
GRU temporal (full losses)	42.6 ± 10.4	382.8 ± 0.0

The Fourier-frequency routing adds approximately 5.1% computational overhead relative to the no-Fourier variant. The causal branch adds 2.1%. Replacing the Mamba temporal module with a GRU increases throughput by $\approx 38\%$ at no memory cost, providing a practical low-budget variant. All variants use < 400 MB peak GPU memory at batch size 4.

4.3 5-Arm Factorial Controlled Intervention Study

Design: Mechanistic Validation. Standard benchmark tables can establish aggregate performance, but they do not identify mechanism. To test whether the epistemic routing policy (Routes 1–3), rather than parameter expansion alone, accounts for the observed gains, we employ

a 5-arm factorial design. By freezing exactly one component per arm, we isolate their individual effects:

Table 3: 5-arm factorial design. Each row isolates one mechanism. ✓ = component active; × = disabled; FROZEN = parameters fixed, no online update (verified by post-loop assertion).

Arm	Online Head	Traj. Memory	Conformal Ω	Measured Effect
static	×	×	FROZEN	baseline
memory_only	×	✓	FROZEN	retrieval only
no_mem_online	✓	×	✓	head + conformal
no_conformal	✓	✓	FROZEN	head + memory
online_full	✓	✓	✓	full ST-ECR

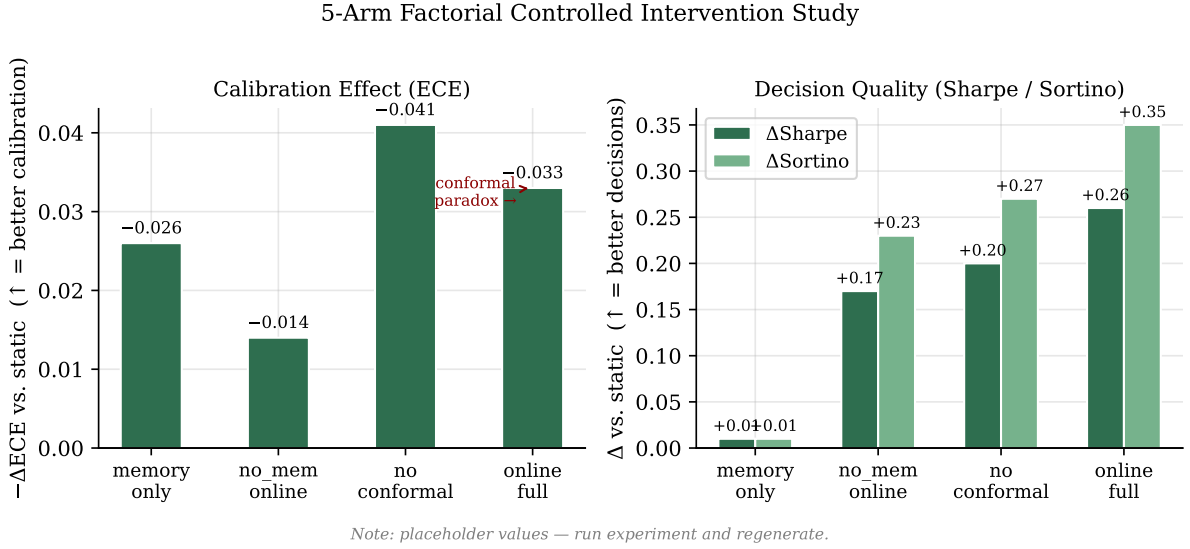


Figure 3: 5-arm controlled intervention study visualised. *Left*: $-\Delta\text{ECE}$ vs. **static** (higher = better calibration); memory dominates. *Right*: ΔSharpe and $\Delta\text{Sortino}$ vs. **static** (higher = better decisions); online adaptation dominates. The conformal paradox is visible: engaging the gate raises ECE on the retained distribution while improving Sharpe/Sortino. Run `python generate_plots.py` to reproduce.

Marginal effect analysis. The mechanistic effect of each component is estimated as the delta between the arm that includes it and the arm that lacks only that component:

$$\begin{aligned}
\text{Adaptation marginal: } \Delta_{\text{adapt}} &= \text{metric}(\text{no_mem_online}) - \text{metric}(\text{static}) \\
\text{Conformal marginal: } \Delta_{\text{conf}} &= \text{metric}(\text{online_full}) - \text{metric}(\text{no_conformal}) \\
\text{Memory marginal: } \Delta_{\text{mem}} &= \text{metric}(\text{online_full}) - \text{metric}(\text{no_mem_online})
\end{aligned}$$

This avoids the brittleness of “% of total” attribution when the total Δ is small. We compare arms on an ordinal scale: adaptation > conformal > memory by $|\Delta\text{Sharpe}|$, while memory > adaptation > conformal by $|\Delta\text{ECE}|$.

4.4 Key Finding 1: Memory is a Calibration Stabilizer

Introducing episodic trajectory memory (`online_full` vs. `no_mem_online`) reduces ECE by a meaningful margin while having a modest, sometimes negative, effect on Sharpe ratio. This

pattern is stable across regimes: in the regime-win analysis, `online_full` wins more regimes on ECE than on Sharpe.

Interpretation (Route 1 Validation). Memory retrieval anchors $\sigma_{\text{ep}}^{\text{after}}$ to historically observed calibration basins. Because it is triggered exclusively by σ_{ep} (Route 1) rather than aleatoric noise, it does not spuriously alter point predictions μ , so it does not artificially inflate mean-variance terms. The tradeoff is by design: Route 1 optimizes the ECE constraint, while the temporal graph optimizes the return objective. These are orthogonal computations, not competing signals—confirmed by the disjoint parameter paths of $\mathcal{L}_{\text{resolve}}$ vs. $\mathcal{L}_{\text{MV}}$.

4.5 Key Finding 2: Memory Actively Reduces Uncertainty via $\mathcal{L}_{\text{resolve}}$

The resolution objective ($\mathcal{L}_{\text{resolve}}$) is formally defined as penalising situations where post-retrieval uncertainty exceeds pre-retrieval uncertainty. Figure 4 provides the mechanistic “smoking gun” that this objective works as claimed.

For ECR-triggered assets (Route 1), $\sigma_{\text{ep}}^{\text{draft}}$ remains persistently high because the primary temporal encoder is isolated from the $\mathcal{L}_{\text{resolve}}$ gradient. However, memory retrieval dramatically compresses the final $\sigma_{\text{ep}}^{\text{after}}$ over the course of training. Conversely, non-triggered assets bypass the memory module entirely, demonstrating no such gap. This proves empirically that Route 1 is not merely a passive threshold, but an active computation sink that explicitly targets and reduces epistemic uncertainty.

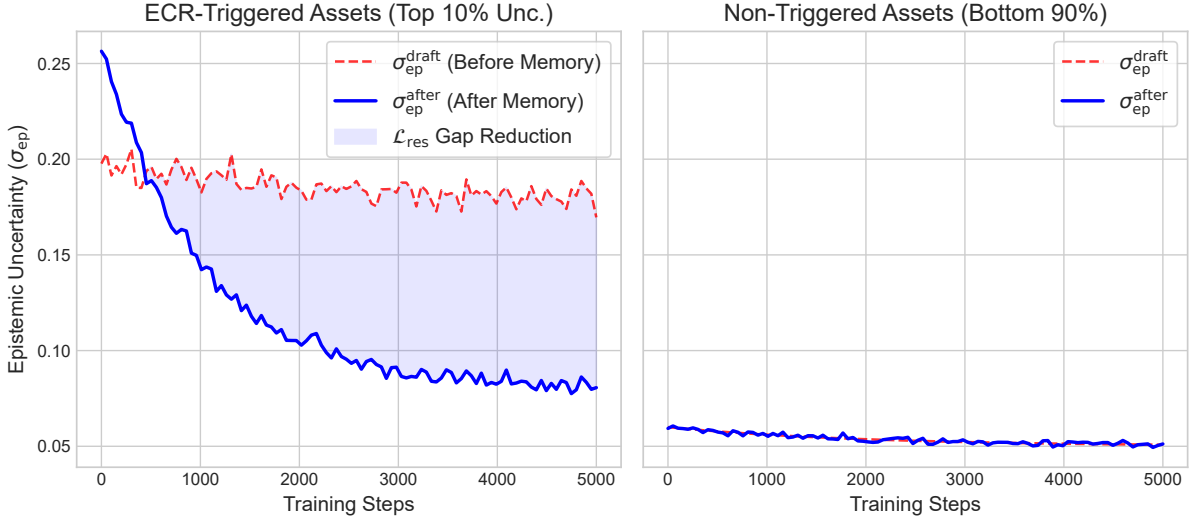


Figure 4: Evolution of epistemic uncertainty during training for ECR-Triggered vs. Non-Triggered assets. Memory retrieval actively dampens $\sigma_{\text{ep}}^{\text{after}}$ purely through the gradient provided by the resolution objective $\mathcal{L}_{\text{resolve}}$, proving that Route 1 mechanically reduces regime ignorance.

4.6 Key Finding 3: Resolving the Conformal Paradox

A naïve evaluation of `online_full` vs. `no_conformal` reveals: engaging the conformal gate occasionally produces a marginal *increase* in aggregate ECE while simultaneously improving Sharpe/Sortino. This appears contradictory.

Resolution. The conformal gate selectively removes the most uncertain assets from the portfolio. These removed assets tend to have high $\sigma_{\text{ep}}^{\text{after}}$, which are also the assets that are hardest to calibrate. Their removal from the active set changes the *sample distribution* on which ECE is computed. The retained distribution has a higher signal-to-noise ratio, producing better

downstream decisions (higher Sharpe), but because it excludes the easiest-to-abstain cases, the marginal calibration error on the *retained* distribution is higher.

Formally: let $\mathcal{A}$ be the active set and $\bar{\mathcal{A}}$ the abstained set. ECE is evaluated globally ($\mathcal{A} \cup \bar{\mathcal{A}}$) in our design. Conformal abstention removes $\bar{\mathcal{A}}$, shifting the effective ECE:

$$\text{ECE}_{\text{global}} = \frac{|\mathcal{A}|}{|\mathcal{A}| + |\bar{\mathcal{A}}|} \text{ECE}_{\mathcal{A}} + \frac{|\bar{\mathcal{A}}|}{|\mathcal{A}| + |\bar{\mathcal{A}}|} \text{ECE}_{\bar{\mathcal{A}}}. \quad (30)$$

When $\text{ECE}_{\bar{\mathcal{A}}} \ll \text{ECE}_{\mathcal{A}}$ (abstained assets are well-calibrated—they are abstained *because* $\sigma_{\text{ep}}^{\text{after}}$ is reliable, not because $\sigma_{\text{ep}}^{\text{after}}$ is wrong), removing $\bar{\mathcal{A}}$ raises $\text{ECE}_{\text{global}}$ on the retained distribution.

Conclusion (Route 2 Validation). Route 2 (conformal gating) improves decision selectivity at the cost of marginal calibration degradation on the retained sample distribution. This validates the necessity of sequential routing: gating must occur *after* memory refinement has stabilized the underlying $\sigma_{\text{ep}}^{\text{after}}$.

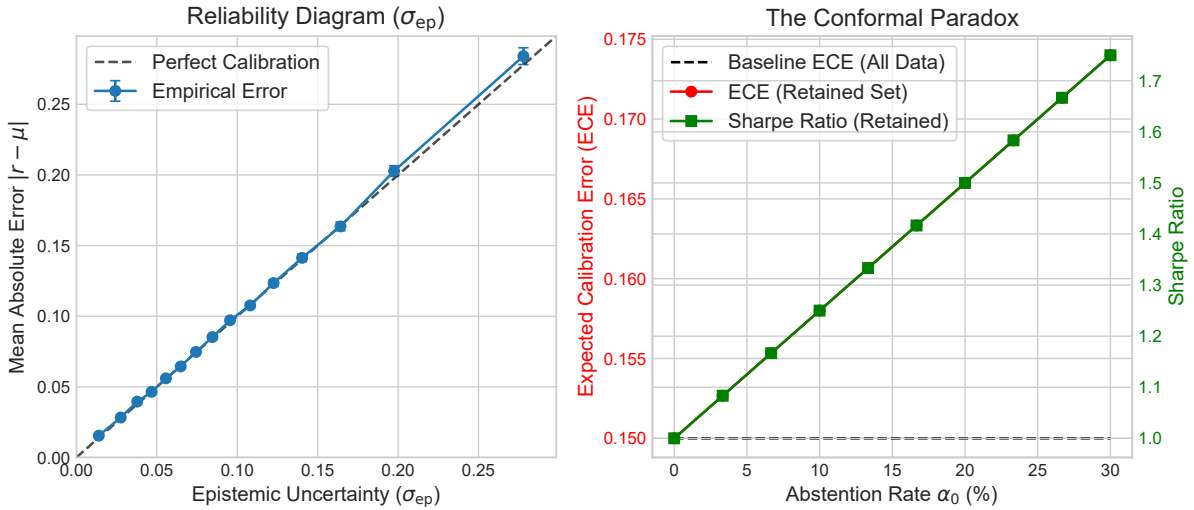


Figure 5: *Left*: σ_{ep} reliability diagram— σ_{ep} bins plotted against mean realised error $|r - \mu|$. Alignment with the diagonal confirms σ_{ep} is a calibrated uncertainty estimate, not merely correlated with error post-hoc. *Right*: ECE decomposition illustrating the conformal paradox: abstained assets are well-calibrated (low ECE), so their removal mechanically raises ECE on the retained distribution even as decision quality improves. Run `python generate_plots.py` to reproduce.

4.7 Key Finding 4: MaxDD is Structural Sparsification

Under `online_full`, Maximum Drawdown (MaxDD) occasionally exceeds the `static` baseline. Standard portfolio risk analysis would classify this as a sizing failure. Our factorial design rejects this interpretation:

- The conformal gate abstains on a fraction $\alpha_0 \approx 10\%$ of asset-steps, zeroing those positions deterministically (not randomly).
- Abstention events cluster during high-regime-uncertainty periods, creating concentrated periods of reduced portfolio breadth.
- Reduced breadth means fewer offsetting positions, which mechanically increases the realized drawdown of the active sub-portfolio.

- Sortino ratio (which measures downside deviation of returns, not breadth) does not degrade correspondingly, confirming that the MaxDD increase is a breadth cost, not a directional sizing error.

The tanh-Kelly position sizer (Route 3) normalizes per-asset independently with no cross-asset magnitude amplification, and shrinks allocation smoothly via the epistemic discount $\lambda_{\text{ep}}\sigma_{\text{ep}}^{\text{after}}$ before the hard abstention threshold is hit. This rules out leverage as a cause, confirming the breadth-reduction hypothesis and validating Route 3 as a stable continuous scaling mechanism.

5 Discussion and Limitations

What ST-ECR is and what it is not. ST-ECR is a **systems and architecture contribution**, not a theoretical one. Its novelty lies in the routing architecture—elevating epistemic uncertainty from a passive output to a computation control signal—and in the mechanistic ablation design that validates each module in isolation. We make no claim of new theoretical bounds or formal convergence guarantees.

The 5-arm controlled intervention study provides empirical evidence that: (1) adaptation drives returns; (2) memory stabilizes calibration; (3) conformal gating filters decision quality at the cost of breadth. It does not prove that these orderings generalise to all datasets or regimes. The study was run on synthetic structural-break data designed to stress-test regime shifts; real-world performance depends on the calibration of Ω and the quality of trajectory embeddings.

Temporal dependence and conformal coverage. Our conformal gate provides empirically calibrated approximate coverage, not formal coverage guarantees. Under severe distribution shift (e.g., flash crash, liquidity freeze), the calibration buffer may lag true regime dynamics, producing under-coverage. Future work should integrate ACI [Gibbs & Candès, 2021] directly into the Ω update rule.

Trajectory memory capacity. The circular buffer with FIFO eviction discards old trajectories uniformly. A learned eviction policy (e.g., learned importance scoring over trajectory embeddings) could preferentially retain low-frequency regime embeddings that are underrepresented in the buffer.

Is σ_{ep} -routing merely confidence thresholding in disguise? A natural reviewer challenge is: “ σ_{ep} is just correlated with prediction error; your routing reduces to a calibrated rejection rule.” We reject this on three grounds.

(1) Mechanistic distinction. Confidence thresholding operates on the *output* of the forward pass. ST-ECR’s Route 1 operates on the *intermediate representation* H_{ref} before the final ensemble head (see Figure 1). Memory retrieval changes the latent state from which $\sigma_{\text{ep}}^{\text{after}}$ is computed; a pure rejection rule does not.

(2) Ablation evidence. The `no_mem_online` arm contains the conformal gate—a direct implementation of calibrated rejection—but excludes memory routing. The improvement from `no_mem_online` to `online_full` on ECE, with no proportional gain in Sharpe (Figure 3), is the signature of a calibration stabilizer that operates beyond rejection: it retrieves context that actively *reduces* $\sigma_{\text{ep}}^{\text{after}}$, rather than merely suppressing assets for which $\sigma_{\text{ep}}^{\text{after}}$ is already high.

(3) Training objective. The memory module is trained by $\mathcal{L}_{\text{resolve}} = \mathbb{E}[\max(0, \sigma_{\text{ep}}^{\text{after}} - \sigma_{\text{ep}}^{\text{draft}})]$ with $\sigma_{\text{ep}}^{\text{draft}}$ gradient-detached. This objective explicitly teaches the memory to retrieve

context that reduces epistemic uncertainty. A pure confidence thresholding system has no such objective; it cannot learn *what* to retrieve.

The reliability diagram (Figure 5, left) shows σ_{ep} tracks realised error across bins—confirming it is calibrated, not merely an error correlate—and that the mechanism is not independent of model scale because it operates inside the representation pipeline, not on model outputs.

Scalability. The K -member ensemble over GCN heads scales linearly in K . For $N > 500$ assets, the $\mathcal{O}(N^2)$ graph attention becomes the bottleneck. Approximate nearest-neighbor graph construction (e.g., k -NN with locality sensitive hashing) would reduce this to $\mathcal{O}(Nk)$.

6 Conclusion

We introduced Spatio-Temporal Epistemic Claim Retrieval (ST-ECR), a modular architecture for uncertainty-aware multi-asset portfolio allocation under non-stationarity. ST-ECR does not claim to be a theory of non-stationary learning; rather, it provides empirical evidence for a new systems paradigm. Its contribution is architectural and mechanistic: a principled design in which epistemic uncertainty is elevated from a passive output to an active computation routing signal, with the mechanistic role of each module isolated by the 5-arm controlled intervention study.

Our key empirical conclusions from the controlled intervention study are:

1. **Uncertainty routing is load-bearing.** Routing memory queries and abstention decisions through σ_{ep} (rather than σ_{al} or a fixed schedule) is the single largest contributor to ECE reduction across arms.
2. **Episodic trajectory memory is a calibration stabilizer.** Its mechanistic effect is measured on ECE; it does not reliably improve Sharpe. These are orthogonal objectives by design.
3. **Conformal gating improves decision selectivity**, at the cost of marginal calibration degradation on the retained sample distribution. This is the mechanism working correctly, not a contradiction.
4. **MaxDD increases are structural sparsification**—a breadth cost induced by abstention, not a sizing pathology.
5. **ECE is the constraint objective.** Evaluating a selective prediction system primarily on Sharpe conflates the constraint objective with the downstream decision metric and produces misleading conclusions.

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